SQL Views & Stored Procedures - Beginner Study Guide

1. Views

A View is a virtual table created from a SELECT query. It stores the query, not the data. It is mainly used to simplify queries, hide sensitive columns, and provide reusable access to data.

Syntax

```
CREATE VIEW vw_EmployeeDetails
AS
SELECT EmpID, EmpName, Department
FROM Employee;
```

Using a View

```
SELECT * FROM vw_EmployeeDetails;
```

Updating through a simple View

```
UPDATE vw_EmployeeDetails
SET Department='IT'
WHERE EmpID=1;
```

Modify a View

```
ALTER VIEW vw_EmployeeDetails
AS
SELECT EmpID, EmpName, Department, Salary
FROM Employee;
```

Delete a View

```
DROP VIEW vw_EmployeeDetails;
```

Benefits of Views

• Hide sensitive columns

• Simplify complex queries

• Reuse SQL logic

• Provide an additional security layer

• Present data like a virtual table

2. Stored Procedures

A Stored Procedure is a reusable database object that stores one or more SQL statements. Procedures can contain business logic and can perform SELECT, INSERT, UPDATE and DELETE operations.

Syntax

```
CREATE PROCEDURE GetEmployees
AS
BEGIN
SELECT * FROM Employee;
END;
```

Execute

```
EXEC GetEmployees;
-- or
EXECUTE GetEmployees;
```

3. Parameterized Stored Procedures

A parameterized stored procedure accepts values from the user at runtime, allowing one procedure to work for many different inputs.

Single Parameter Example

```
CREATE PROCEDURE GetEmployeesByDepartment
@Dept VARCHAR(30)
AS
BEGIN
SELECT *
FROM Employee
WHERE Department=@Dept;
END;
```

Execute:

```
EXEC GetEmployeesByDepartment @Dept='IT';
EXEC GetEmployeesByDepartment @Dept='HR';
```

Multiple Parameters Example

```
CREATE PROCEDURE GetEmployeeByDeptAndSalary
@Dept VARCHAR(30),
@MinSalary DECIMAL(10,2)
AS
BEGIN
SELECT *
FROM Employee
WHERE Department=@Dept
AND Salary>@MinSalary;
END;
```

Execute:

```
EXEC GetEmployeeByDeptAndSalary
@Dept='IT',
@MinSalary=60000;
```

Views vs Stored Procedures

Feature	View	Stored Procedure
Stores Data?	No	No

Stores SQL?	SELECT query	One or more SQL statements
Accepts Parameters?	No	Yes
Can INSERT/UPDATE/DELETE?	Limited (simple views)	Yes
Purpose	Virtual table	Reusable business logic

Interview Questions

• What is a View?

• What is a Stored Procedure?

• Difference between a View and a Stored Procedure?

• What is a parameterized stored procedure?

• Can a View accept parameters?

• Can a stored procedure perform UPDATE or DELETE operations?